

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: As-73			FORMS: Soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 80.3 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
11	12	42*	76		
53	10	12*	51		
		13*	10		
		53*	10		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 20 mCi B – level (moderate hazard): > 20 mCi, < 1 Ci A – level (high hazard): > 1 Ci					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 0.056 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 0 mrem/h Contamination skin dose, uniform deposit of 1 μ Ci per cm ² : 17 mrem/h			INTERNAL RADIATION HAZARDS Annual Limit on Intake: 8000 μCi (Ingestion) 2000 μCi (Inhalation) The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING Gammas/X-rays: 0.319 mm of lead will reduce the gamma dose rate by 90%. Betas/electrons: <0.1 mm of plastic will absorb all emissions. Bremsstrahlung may be created and require additional shielding.			DOSIMETRY AND BIOASSAY REQS Urine assays may be required after large spills or contaminations.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: Thin-window NaI. 2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 3. Survey work areas before, during, and after work. Instrument and smear surveys are required. 4. Segregate waste to those with half-lives between 15 and 120 days. 5. Limit soluble waste to the sewer to less than 100 μCi/day per lab.					

References:

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- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>